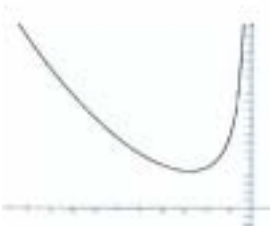
Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)	184	**2**	**M1** for $\frac{852-300}{300}[\times 100]$ oe or for $\frac{852}{300}\times 100[-100]$ oe
1(b)	497	**2**	**M1** for $\frac{852}{5+7}\times k$ oe where $k = 1$, 5 or 7
1(c)(i)	Forty thousand six hundred	**1**	
1(c)(ii)	4.06×10^4	**1**	
1(d)	435	**3**	**M2** for $3000\times\left(1-\frac{48}{100}-\frac{3}{8}\right)$ oe or **B2** for 2565, or 1440 **and** 1125 or 1875 **and** 1440 or 1560 **and** 1125 or **M1** for $1-\frac{48}{100}-\frac{3}{8}$ or $3000\times\left(\frac{48}{100}+\frac{3}{8}\right)$ oe or **B1** for 1440 or 1125 or 1560 or 1875 If 0 scored **SC1** for answer 975
1(e)	35.7	**3**	**M2** for $\frac{100+15}{100}\times\frac{100+18}{100}$ [–1] oe or better or **M1** for $k\times\frac{100+15}{100}\times\frac{100+18}{100}$ oe
2(a)	1[.0] 0.9	**2**	**B1** for each
2(b)	correct curve 	**4**	**B3 FT** for 6 or 7 points **B2 FT** for 4 or 5 points **B1 FT** for 2 or 3 points

Question	Answer	Marks	Partial Marks
2(c)	ruled line at $y = -1$	**B1**	
	0.3 to 0.32	**B1**	
3(a)(i)	169	**2**	**M1** for g(13) or $(1+4x)^2$ or better
3(a)(ii)	$1+4x^2$ final answer	**1**	
3(a)(iii)	x	**1**	
3(b)	3.5 or $\frac{7}{2}$	**2**	**M1** for $1+4x=15$
4(a)(i)	40.9 or 40.91…	**3**	**M2** for [sin ABC=] $\frac{29.5\sin 51.6}{35.3}$ oe or for [cos ABC =] $\frac{35.3^2+45^2-29.5^2}{2\times 35.3\times 45}$ or **M1** for $\frac{29.5}{\sin ABC}=\frac{35.3}{\sin 51.6}$ oe or for correct implicit cosine rule
4(a)(ii)	520 or 520.0 to 520.2…	**2**	**FT** *their* (a)(i) if used provided working shown **M1** for $0.5\times 29.5\times 45\times \sin 51.6$ oe or for $0.5\times 35.3\times 45\times \sin(\textit{their}(\text{a})(\text{i}))$ or for 0.5 × 35.3 × 29.5sin (180–51.6–*their*(a)(i))
4(b)(i)	41.2 or 41.21 to 41.23	**4**	**M1** for $SQ = 2\times 32\times \sin\left(\frac{1}{2}\times 56\right)$ oe or $\sqrt{32^2+32^2-2\times 32\times 32\times \cos 56}$ oe or $\frac{32\sin 56}{\sin((180-56)\div 2)}$ oe **M2** for $SR^2 = 47^2+\left(\textit{their } SQ^2\right)-2\times 47\times \textit{their } SQ\times \cos 60$ or **M1** for implicit form
4(b)(ii)	28.3 or 28.25 to 28.29…	**3**	**M2** for 32 × sin62 oe or **M1** for recognition that line from P is perpendicular to SQ
5(a)	121 or 120.8… or 120 $\frac{5}{6}$	**4**	**M1** for midpoints soi **M1** for use of $\sum fx$ with x in correct interval including both boundaries but **not** if x is 50, 50, 100 and 300 **M1** (dep on 2nd M1) for $\sum fx \div 120$